

# An Overview of the 4th International Workshop on eXtended Reality for Industrial and Occupational Supports (XRIOS)

Bernardo Marques <sup>\*1</sup>, Samuel Silva <sup>†1</sup>, Fábio Barros <sup>‡1</sup>, Tiago Araújo <sup>§2</sup>,  
Carlos Ferreira <sup>¶3</sup>, Paulo Dias <sup>||1</sup>, Beatriz Sousa Santos <sup>\*\*</sup>

<sup>1</sup> IEETA, DETI, LASI, University of Aveiro, Portugal; <sup>2</sup> IEETA, ESAN, LASI, University of Aveiro, Portugal;

<sup>3</sup> IEETA, DEGEIT, LASI, University of Aveiro, Portugal

## ABSTRACT

The 4th International Workshop on the eXtended Reality for Industrial and Occupational Supports (XRIOS) aims to identify the current state of XR research and the gaps in the scope of human factors and ergonomics, related to industrial and occupational tasks, and discuss potential future directions. XRIOS was held for the first time at IEEE VR 2022, where it served as the first venue to build an interdisciplinary community that bridges XR developers/practitioners and human factors and ergonomics researchers interested in industrial and occupational XR applications. XRIOS 2025 follows the success of previous editions in response to society's growing needs by expanding the community and providing more opportunities.

**Keywords:** eXtended Reality, Virtual Reality, Augmented Reality, Industrial Support, Occupational Support, Human Factors.

## 1 OVERVIEW

Technologies exploring eXtended Reality (XR), encompassing virtual, augmented, and mixed reality, are increasingly being explored to support industrial and occupational tasks by enhancing productivity, safety, and training [2, 7]. Solutions using XR enable immersive and interactive environments for workforce training, allowing users to practice complex tasks in realistic simulations without real-world risks. It also aids in improving operational efficiency through augmented visualizations of data, real-time instructions, and remote collaboration. In industrial settings, XR facilitates predictive maintenance, task planning, or inspection processes, among others by integrating digital twins and advanced sensing technologies [1, 3, 5]. As XR continues to evolve, its potential to address challenges related to ergonomics, human-machine interaction, or occupational safety makes it a transformative technology for various use-cases [4, 6].

This workshop aims to explore the current state of XR research, identifying gaps in human factors and ergonomics with a focus on industrial and occupational tasks, while fostering discussions on future research directions. It seeks to build a collaborative community by connecting XR developers/practitioners with researchers. Additionally, it provides a platform for academic and industry professionals to showcase their latest work or ongoing research.

\*e-mail: bernardo.marques@ua.pt

†e-mail: sss@ua.pt

‡e-mail: fabiodaniel@ua.pt

§e-mail: tiagodavi70@ua.pt

¶e-mail: carlosf@ua.pt

||e-mail: paulo.dias@ua.pt

\*\*e-mail: bss@ua.pt

## 2 CALL FOR PAPERS

XRIOS 2025<sup>1</sup> is held at IEEE VR 2025. We will have multiple research paper presentations, and discussion sessions, as well as an invited talk. Here, we describe our approach to promote paper submissions with the details of paper topics and types.

### 2.1 Topics

To cover the wide range of XR applications in the context of industrial and occupational supports, we tried to include various XR-related topics, which embrace different disciplines and perspectives. Some of the highlighted topic areas include, but are not restricted to:

- Industrial and occupational applications in XR;
- XR in various dynamic workplaces (e.g., construction, health-care, transportation, etc.);
- XR for industrial job training and hygiene;
- Exoskeletons with XR for rehabilitation;
- Human-robot interaction/collaboration with XR;
- Physiological and fatigue assessment in XR;
- Physical and cognitive workload in XR;
- Ergonomic considerations for wearable XR devices;
- Remote collaboration and learning in XR;
- Office/workplace ergonomics in XR;
- Measures and human-performance modeling in XR;
- Digital twin for industrial applications.

### 2.2 Paper Types and Advertisement

While inviting the community, we sought various types of paper submissions (up to 8 pages, excluding references), which include:

- **Position Papers** identifying insightful opinions and ideas;
- **Survey Papers** capturing the current states of research and presenting potential research gaps and future directions;
- **Project Papers** providing an overview of an ongoing/planning project by describing the approach and goals of the project;
- **Research Papers** describing state-of-the-art advances and providing results as evidence, which can include industrial/occupational XR applications and systems.

For effective communication and visibility, we designed a digital call-for-paper (CfP) poster. The CfP flyer was shared in the relevant research communities via multiple online resources (e.g., mailing lists), and social media (e.g., LinkedIn, Facebook, Slack).

### 2.3 Important Dates

All deadlines were 23:59:59 Anywhere on Earth (GMT/UTC-12:00).

- **Paper Submission:** December 22, 2024
- **Acceptance Notification:** January 8, 2025
- **Camera-ready for inclusion in the IEEE Digital Library:** January 15, 2025
- **XRIOS Workshop:** March 9, 2025

<sup>1</sup><https://sites.google.com/view/xrios>

## 2.4 XRIOS Team

We formed a collaborative group of fifteen international researchers with intensive research expertise in XR and human factors for organizing this workshop. Besides the authors of this paper, from the University of Aveiro, our International Program Committee had the collaboration of: Hyungil Kim - University of Illinois Chicago, USA; Isaac Cho - Utah State University, USA; Kangsoo Kim - University of Calgary, Canada; Dongyun Han - Utah State University, USA; Allison Bayro - Arizona State University, USA; Heejin Jeong - Arizona State University, USA; Hayoun Moon - Virginia Tech, USA; Myounghoon Jeon - Virginia Tech, USA.

## 2.5 Best Paper Awards

A few high-quality papers (e.g., Best Papers) will be selected and invited to submit their extended papers to Elsevier Computers & Graphics Journal<sup>2</sup> - "Special Section on Recent Advances in Industrial eXtended Reality (XR)" (Guest Editors: Bernardo Marques, Samuel Silva, Hyungil Kim).

## 3 ORGANIZATION AND SCHEDULE

The tentative schedule will include one invited talk and 13 paper presentations. Small group discussions may also occur if possible.

### 3.1 Invited Talk

- **Title:** Multimodal Interactions for Remote Collaboration and Training in XR
- **Abstract:** Professor Joaquim Jorge will discuss the role of multimodal interaction systems in enhancing remote collaboration and training in industrial settings, focusing on AI-enhanced XR. These systems provide immersive training tools in virtual environments by integrating eye-gaze tracking, body gestures, foot input, and haptic feedback. AI's role in interpreting user actions for precise guidance and real-time feedback will be highlighted and insights will be drawn. The talk will address how these systems support remote team coordination, improve training outcomes, and maintain user engagement during extended sessions. Practical applications in maintenance and operational training, and the exciting future directions for more adaptive XR training platforms, which promise to revolutionize how we train and collaborate.

### 3.2 A Summary of Accepted Papers

A total of 13 papers were accepted for presentation. All submissions were reviewed using a single-blind peer-review process. Each paper was evaluated by two reviewers, ensuring an impartial and thorough assessment while avoiding potential conflicts. Among these, several papers focused on leveraging XR technologies in healthcare, industry, and collaborative environments, reflecting the evolving landscape of XR research and its applications. Below is a thematic summary of the key contributions and trends observed in the accepted papers. Regarding **healthcare applications**, 2 papers addressed stroke rehabilitation. The papers explored VR-based approaches to enhance stroke rehabilitation, focusing on personalized, adaptive systems and multi-user frameworks that emphasize social interaction for improved therapy outcomes. As for **Industry and Maintenance**, 3 papers explored industrial applications of XR technologies. Particularly, for predictive maintenance, task guidance, and teleoperation. These innovations aim to optimize workflows, reduce technical barriers, and enhance precision in industrial environments. Concerning **Training and Simulation**, 2 papers demonstrated the value of immersive XR platforms for situational awareness and skill development, including applications in aviation training and real-time management of critical infrastructure systems. **Collaborative Scenarios** was a central theme in 3 papers. They

examined how XR can improve teamwork and accelerate prototyping processes, with studies showcasing AR tools for collaboration, rapid prototyping techniques, and effective balancing of digital and physical tools. In respect of **Advanced Interaction Techniques**, 2 papers proposed novel XR frameworks, supporting integration of multimodal features as well as human-centered digital twins for applications like safety training and ergonomics.

The reviews obtained illustrate that the degree of innovation of the papers spans from medium to high, with many papers introducing novel approaches to XR frameworks. Plus, technical complexity is generally rated as medium or high, reflecting advanced algorithms, interfaces, or engineering efforts involved in these studies. As to maturity of the solutions, it varied widely, with several studies presenting medium-maturity prototypes tested in controlled environments. High-maturity papers demonstrate fully realized solutions evaluated in realistic contexts. Most papers (10 out of 13) included user-based experiments, reflecting the growing trend of empirical validation in XR research. All in all, this diverse collection of research contributions underscores the expanding role of XR in healthcare, industry, training, and collaboration, offering a glimpse into the future of immersive technologies.

## 4 CONCLUDING REMARKS

We hope that XRIOS 2025 continues the success of the previous editions. This overview of the accepted papers with diverse concept designs, research methods, and XR application prototypes significantly contributes to the design of XR technologies in various industry settings and guides researchers to more impactful research directions. We will continue this workshop series in the future through IEEE VR and other conferences (e.g., ACM CHI and IEEE ISMAR Conferences). Also, we will guest-edit a special issue on the same topic in Elsevier Computers & Graphics Journal.

## ACKNOWLEDGMENTS

We thank all paper authors who submitted and presented their work at the workshop. Also, the international program committee, Prof. Joaquim Jorge and the IEEE VR 2025 Workshop Chairs. This work was supported by FCT under unit 00127-IEETA.

## REFERENCES

- [1] A. Alhakamy. eXtended Reality (xr) toward building immersive solutions: The key to unlocking industry 4.0. *ACM Computing Surveys*, 56(9):1–38, 2024.
- [2] L. A. Cardenas-Robledo, Ó. Hernández-Urbe, C. Reta, and J. A. Cantoral-Ceballos. eXtended Reality applications in industry 4.0. a systematic literature review. *Telematics and Informatics*, 73, 2022.
- [3] H. Jalo, H. Pirkkalainen, O. Torro, E. Pessot, A. Zangiacomi, and A. Tepljakov. Extended reality technologies in small and medium-sized european industrial companies: level of awareness, diffusion and enablers of adoption. *Virtual Reality*, 26(4):1745–1761, 2022.
- [4] K. Kim, B. Marques, H. Jeong, S. Silva, I. Cho, C. Ferreira, H. Kim, P. Dias, M. Jeon, and B. S. Santos. An overview of the 2nd international workshop on eXtended Reality for industrial and occupational supports (xrios). In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 364–369, 2023.
- [5] R. Maio, B. Marques, A. Santos, P. Ramalho, D. Almeida, P. Dias, and B. S. Santos. Real-time data monitoring of an industry 4.0 assembly line using Pervasive Augmented Reality: First impressions. In *IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 414–417, 2023.
- [6] B. Marques, C. Ferreira, S. Silva, A. Santos, A. Santos, P. Dias, and B. S. Santos. Exploring different content creation and display methods for remote collaboration supported by eXtended Reality: comparative analysis of distinct task scenarios. *Multimedia Tools and Applications*, pp. 1–31, 2024.
- [7] P. A. Rauschnabel, R. Felix, C. Hinsch, H. Shahab, and F. Alt. What is XR? towards a framework for augmented and virtual reality. *Computers in human behavior*, 133, 2022.

<sup>2</sup>[sciencedirect.com/journal/computers-and-graphics](https://www.sciencedirect.com/journal/computers-and-graphics)